CURRENT AND FUTURE GENERATION

World Energy Needs and Nuclear Power

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- The world will need significantly increased energy supply in the future, especially cleanly-generated electricity.
- Electricity demand is increasing about twice as fast as overall energy use and is likely to rise by more than half between 2022 and 2040.
- Nuclear power provides about 10% of the world's electricity, and about 20% of Europe's electricity.
- Almost all reports on future energy supply from major organizations suggest an increasing role for nuclear power as an environmentally benign way of producing reliable electricity on a large scale.

Growth in the world's population and economy, coupled with rapid urbanization, will result in a substantial increase in energy demand over the coming years. The United Nations (UN) estimates that the world's population will grow from about 8 billion in 2024 to around 9.8 billion by 2050. The process of urbanization – which currently adds a city the size of Shanghai to the world's urban population every four months or so – will result in approximately two-thirds of the world's people living in urban areas by 2050 (up from about 55% in 2022). The challenge of meeting rapidly growing energy demand, whilst reducing harmful emissions of greenhouse gases, is considerable. In 2023 global energy-related carbon dioxide (CO_2) emissions rose to 37.4 Gt, the highest on record, and over 60% above the total in 2000 (23.2 Gt).

Privacy - Terms

Electricity demand growth has outpaced growth in final energy demand for many years. Increased electrification of end-uses – such as transport, space cooling, large appliances, and ICT – are key contributors to rising electricity demand. The number of people without access to electricity has fallen substantially, and in 2017 was below one billion. However, despite significant progress, around 9.6% of the world's population – about 760 million people – living mostly in rural areas, were without access in 2022.

Aside from the challenges of meeting increasing demand and reducing greenhouse gas emissions, cleaner air is a vital need. According to the World Health Organization (WHO), air pollution is the world's largest environmental risk. The WHO estimates that about seven million people die prematurely as a result of air pollution. Much of the fine particulate matter in polluted areas arises from industrial sources such as power generation or from indoor air pollution which could be averted by electricity use.

Nuclear energy is a low-emitting source of electricity production and is also specifically low-carbon, emitting among the lowest amount of carbon dioxide equivalent per unit of energy produced when considering total life-cycle emissions. It is the second largest source of low-carbon electricity production globally (after hydropower), and provided about 26% of all low-carbon electricity generated in 2022. Almost all reports on future energy supply from major organizations suggest an expanded role for nuclear power is required, alongside growth in other forms of low-carbon power generation, to create a sustainable future energy system.

In June 2019 the OECD's International Energy Agency (IEA) published a report, Nuclear Power in a Clean Energy System, which concluded that a failure to invest in existing and new nuclear plants in advanced economies would make global efforts to transition to a cleaner energy system drastically harder and more costly.

In June 2022 the IEA report on Nuclear Power and Secure Energy Transitions concluded that nuclear energy can "help make the energy sector's journey away from unabated fossil fuels faster and more secure," with nuclear being "well placed to help decarbonize electricity supply." The report emphasizes the significant role nuclear plants can play in securing the global pathway to net zero carbon emissions.

Primary energy and electricity outlook

There are many outlooks for primary energy and electricity published each year, many of which are summarized below. Among the most widely-referenced organizations in this regard is the IEA.

Several scenarios are produced and regularly updated by the IEA to explore different versions of the future and the circumstances that could lead to them. Two main approaches are used to develop these scenarios:

- Exploratory – this method applies the effects of certain policies and actions being carried out to build up possible versions of the future.
- Normative – derives potential pathways needed to achieve a particular outcome by a given date.

The former approach (exploratory) provides a picture of what the future might look like if certain conditions are met. Usually a 'business-as-usual' scenario would be provided as a reference point to show what the outcome could be if current policies and plans were to continue their trend through the forecast period. The IEA's Stated Policies Scenario (STEPS) – which is "designed to provide a sense of the prevailing direction of energy system progression, based on a detailed review of the current policy landscape" – is an example of this type of scenario.

In the **WEO 2023** Stated Policies Scenario (STEPS), global energy needs rise by about 21% to 2050, and global electricity demand nearly doubles. Growth in demand comes largely from emerging markets and developing economies. Almost all net growth in demand is met by low emissions sources, but annual emissions are only reduced by about 20%.

In the STEPS scenario, China's energy demand reduces slightly between 2030 and 2050, but in 2050 still accounts for 20% of the world total.

There are many changes ahead in the sources of primary energy used. The dominance of fossil fuels is reduced modestly across the scenarios, declining from 79% of total primary supply in 2022, to 60% by 2050 in the STEPS and 16% in the Net Zero Emissions by 2050 (NZE) Scenario. Despite the relative decrease, the absolute amount of energy consumed either directly or indirectly through the burning of fossil fuels decreases by just 14% to 2050 in the STEPS, and by about 82% in the NZE Scenario. The proportion of final energy consumption that is in the form of electricity increases from

20% in 2022, to 30% by 2050 in the STEPS, and to 53% in the NZE Scenario.

As the use of electricity grows significantly, the primary energy sources used to generate it are changing. In 2022, 61% of the electricity generated globally was through the burning of fossil fuels. Whilst the STEPS sees this figure reduced to 21% of the total, absolute electricity generation in 2050 from fossil fuels remains at 65% of 2022 levels. The Net Zero Emissions by 2050 Scenario sees the fossil fuel share of generation markedly reduced to just 1% of total generation by 2050, with absolute generation 7% of that in 2022. In both scenarios, generation from all low-carbon sources of electricity is required to grow substantially.

Nuclear power for electricity in published scenarios

Nuclear power generation is an established part of the world's electricity mix providing about 10% of world electricity. It is especially suitable for meeting large-scale, continuous electricity demand where reliability and predictability are vital – hence ideally matched to increasing urbanization worldwide.

Almost all reports on future energy supply from major organizations suggest an increasing role for nuclear power as an environmentally benign way of producing reliable electricity on a large scale.

The energy projections produced by the IEA in particular are frequently consulted by policymakers, the media, and analysts. The nuclear power sector projections of the main IEA scenarios, alongside those produced by the International Atomic Energy Agency and World Nuclear Association, are discussed in detail in the information page on IEA Scenarios and the Outlook for Nuclear Power.

MIT Future of Nuclear Energy in a Carbon-Constrained World

A major two-year study by the Massachusetts Institute of Technology Energy Initiative (MITEI) published in September 2018 underlined the pressing need to increase nuclear power generation worldwide. It outlined measures to achieve this, including moves to reduce the cost of building new nuclear capacity and creating a level playing field that would allow all low-carbon generation

technologies to compete on their merits. "While a variety of low- or zero-carbon technologies can be employed in various combinations, our analysis shows the potential contribution nuclear can make as a dispatchable low-carbon technology. Without that contribution, the cost of achieving deep decarbonisation targets increases significantly," the study finds. The MIT study is designed to serve as a balanced, fact-based, and analysis-driven guide for stakeholders involved in nuclear energy, notably governments.

With high carbon constraints, the system costs of electricity without nuclear power is twice as high in the USA and four times as high in China according to the MIT study.* Scenarios envisage nuclear comprising over half of capacity in the USA and over 60% in China if overall carbon emissions are reduced to 50 g/kWh.

* Nominal overnight capital cost of nuclear is $5500/kW in the USA and $2800/kW in China, possibly reducing to $4100/kW and $2100/kW.

International Atomic Energy Agency

The 43rd edition of the IAEA's annual Reference Data Series No. 1 report, titled Energy, Electricity and Nuclear Power Estimates for the Period up to 2050, contains estimates of energy, electricity and nuclear power trends up to the year 2050. The low case is essentially a 'business-as-usual' scenario where the current market, technology and resource trends continue. The high case assumptions are more ambitious, with policies on climate change also being taken into account.

The IAEA quotes the 2022 nuclear capacity as 371 GWe from 411 operable reactors; a further 22.8 GWe from 24 units in Japan and four units in India are classed as 'suspended operation'. This would increase to 458 GWe in the low case and 890 GWe in the high case by 2050.

US Energy Information Administration

The US Energy Information Administration (EIA) publishes an annual report called *International Energy Outlook (IEO)*.

In ***IEO 2023***, electricity from renewables is projected to increase by about 240% between 2022 and 2050, accounting for 50% of global electricity generation by 2050. Nuclear generation is projected to increase by 22% during this period, but relative to total

generation, the share of nuclear generation would fall from about 10% of total electricity generation in 2022 to about 8% in 2050.

Institute of Energy Economics, Japan

The Institute of Energy Economics, Japan (IEEJ) Outlook 2024 report shows nuclear energy helping Asian countries achieve future economic growth, energy security and environmental protection. In the Reference scenario, nuclear electricity generation would increase from 2808 TWh to 3511 TWh by 2050 but its share of total global electricity generation would decline to 7.2% from about 10% in 2022.

In the Advanced Technologies Scenario, in which each country "will actively implement aggressive energy efficiency and decarbonisation policies," the IEEJ projects an acceleration of nuclear power plant construction and an improvement in average capacity factor. In this scenario nuclear generation increases to 5565 TWh by 2050, maintaining its share of total of supply at about 10%.

BP

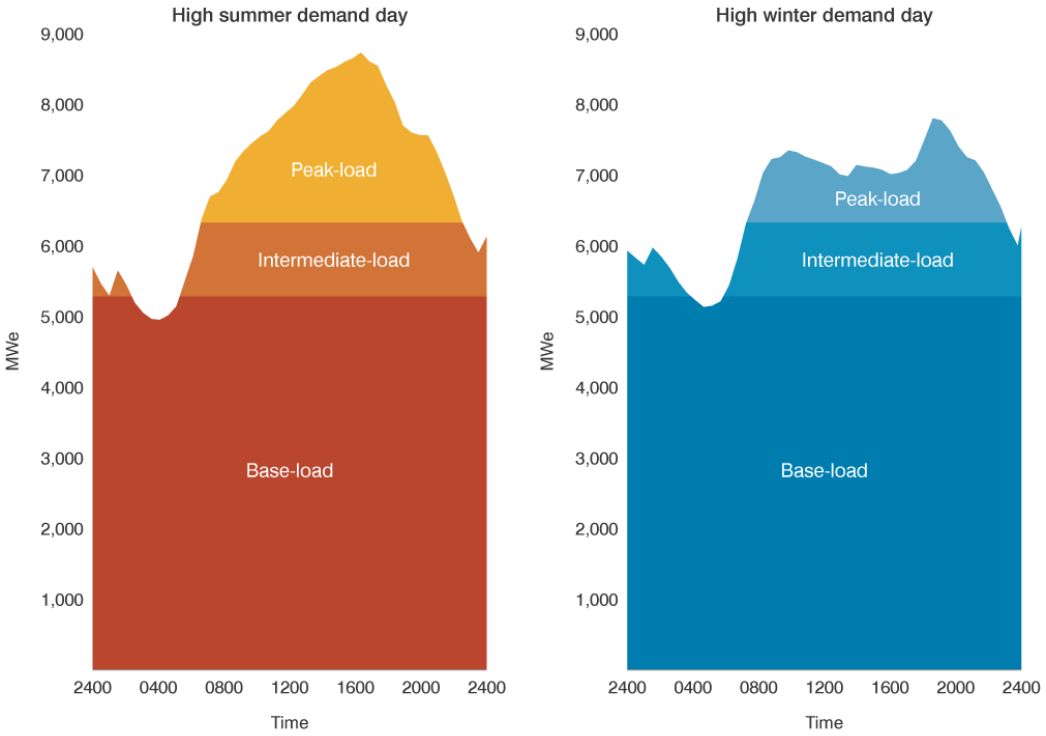
BP's 2024 Energy Outlook includes 'Current Trajectory' and 'Net Zero' scenarios. Modest growth in primary energy consumption of about 7% is expected to 2050 in the Current Trajectory scenario. In the Net Zero scenario primary energy consumption declines by over 25%. In both scenarios supply of energy from nuclear grows significantly. In the Current Trajectory scenario supply increases by 50% to 2050, and in the Net Zero scenario it more than doubles. In both scenarios almost all growth in supply of energy from nuclear is from the Asia Pacific region, principally China.

Generation options

In electricity demand, the need for low-cost continuous, reliable supply can be distinguished from peak demand occurring over a few hours daily and able to command higher prices. Supply needs to match demand instantly and reliably over time. There are a number of characteristics of nuclear power which make it particularly valuable apart from its actual generation cost per unit – MWh or kWh. Fuel is a low proportion of power cost, giving power price stability, its fuel is on site (not depending on continuous

delivery), it is dispatchable on demand, it has fairly quick ramp-up, it contributes to clean air and low-CO_2 objectives, it gives good voltage support for grid stability. These attributes are mostly not monetized in merchant markets, but have great value which is increasingly recognized where dependence on intermittent sources has grown, and governments address long-term reliability and security of supply.

The renewable energy sources for electricity constitute a diverse group, from wind, solar, tidal, and wave energy to hydro, geothermal, and biomass-based power generation. Apart from hydropower in the few places where it is very plentiful, all of the renewables have limitiations, either intrinsically or economically, in potential use for large-scale power generation where continuous, reliable supply is needed.



This diagram shows that much of the electricity demand is in fact for continuous 24/7 supply (base-load), while some is for a lesser amount of predictable supply for about three-quarters of the day, and less still for variable peak demand up to half of the time.

Apart from nuclear power the world relies almost entirely on fossil fuels, especially coal, to meet demand for base-load electricity production. Most of the demand is for continuous, reliable supply on a large scale and there are limits to the extent to which this can be changed.

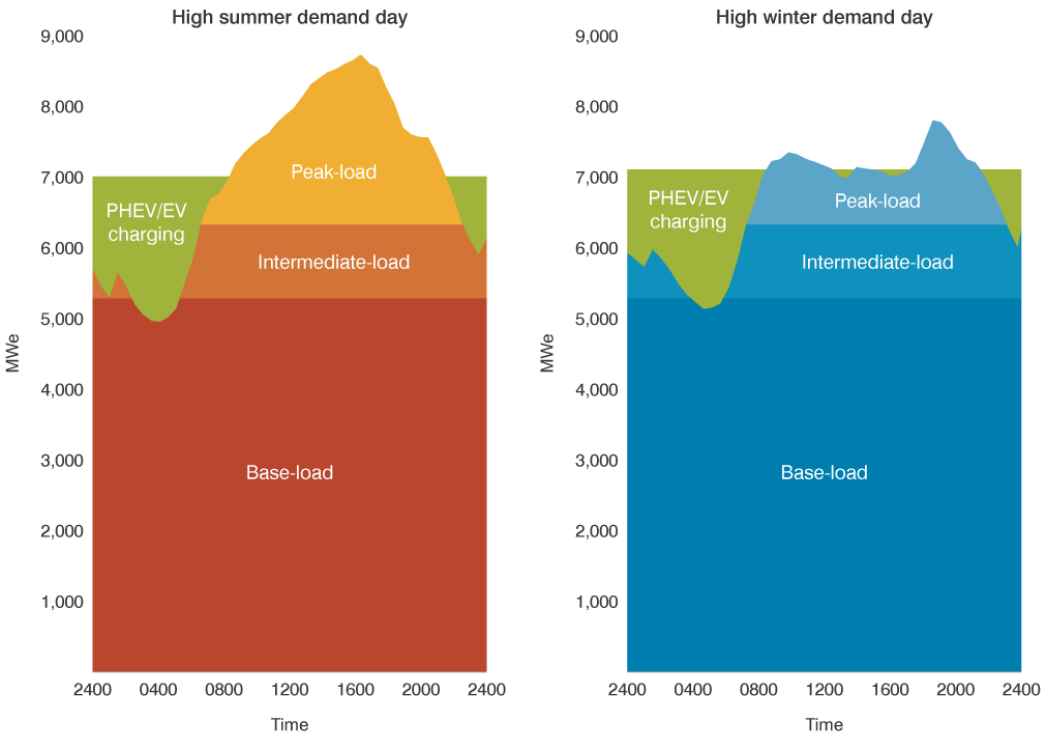
Natural gas is increasingly used as fuel for electricity generation in many countries. The challenges associated with transport over long

distances and storage are to an extent alleviated through liquefaction. However much storage remains underground, in depleted oilfields, especially in the USA, and this can be dangerous. In 2015 the Aliso Canyon storage field in California leaked for some months, releasing about 66 tonnes of methane per hour, causing widespread evacuation and neutralising the state's efforts to curb carbon emissions (methane has over 25 times greater global warming potential than carbon dioxide).

Implications of electric vehicles

Future widespread use of electric vehicles, both pure electric and plug-in hybrids, will increase electricity demand modestly – perhaps up to 15% in terms of kilowatt-hours. But this increase will mostly come overnight, in off-peak demand, so will not significantly increase systems' peak capacity requirement in gigawatts. Overnight charging of vehicles will however greatly increase the proportion of that system capacity to be covered by base-load power generation – either nuclear or coal. In a typical system this might increase from about 50-60% to 70-80% of the total, as shown in the Figures below.

This then has significant implications for the cost of electricity. Base-load power is generated much more cheaply than intermediate- and peak-load power, so the average cost of electricity will be lower than with the present pattern of use. And any such major increase in base-load capacity requirement will have a major upside potential for nuclear power if there are constraints on carbon emissions. So potentially the whole power supply gets a little cheaper and cleaner, and many fossil fuel emissions from road transport are avoided at the same time.



Drivers for increased nuclear capacity

The first generation of nuclear plants were justified by the need to alleviate urban smog caused by coal-fired power plants. Nuclear was also seen as an economic source of base-load electricity which reduced dependence on overseas imports of fossil fuels. Today's drivers for nuclear build have evolved.

Increasing energy demand

Global population growth in combination with industrial development will lead to strong growth in electricity consumption in the decades ahead. Besides the expected incremental growth in demand, there will be the challenge of renewing a lot of existing generating stock in the USA and the EU over the same period. An increasing shortage of fresh water calls for energy-intensive desalination plants.

Climate change and sustainable development

Increased awareness of the dangers and effects of global warming and air pollution has led decision-makers, media, and the public to realize that the use of fossil fuels must be reduced and replaced by low-emissions sources of energy, such as nuclear power – the only readily available large-scale alternative to fossil fuels for production of a continuous, reliable supply of electricity.

For more information see page on Nuclear Energy and Sustainable Development.

Security of supply

A major topic on many political agendas is security of supply, as countries realize how vulnerable they are to interrupted deliveries of oil and gas. The abundance of naturally occurring uranium makes nuclear power attractive from an energy security standpoint.

For more information see page on Nuclear Power and Energy Security.

Economics

As carbon emission reductions are encouraged through various forms of government incentives and trading schemes, the economic benefits of nuclear power will increase further.

For more information see page on Economics of Nuclear Power.

Insurance against future price exposure

A longer-term advantage of uranium over fossil fuels is the low impact that variable fuel prices have on final electricity production costs. This insensitivity to fuel price fluctuations offers a way to stabilize power prices in deregulated markets.

Is a rapid expansion of nuclear power capacity possible?

It is noteworthy that in the 1980s, 218 power reactors started up, an average of one every 17 days. These included 47 in the USA, 42 in France and 18 in Japan. The average power was 923.5 MWe. It is not hard to imagine a similar number being commissioned in the future.

See also the page in this series: Heavy Manufacturing of Power Plants.

Appendix

IEA: World Energy Outlook

Annual editions of WEO from the OECD International Energy Agency (IEA) make clear the increasing importance of electricity, with all scenarios expecting demand growth to outpace that of total final energy demand. Also clear across successive reports is the growing role that nuclear power will play in meeting global energy needs, while achieving security of supply and minimizing carbon dioxide and air pollutant emissions.

WEO 2021 presents electricity generation growth of between 75% and 116% over 2020-2050 across its three main scenarios. In the report's Sustainable Development Scenario, nuclear generation increases by 2022 TWh (75%) over the same period, requiring capacity growth of about 254 GW, or 61%.

WEO 2020 presents electricity generation growth of between 46% and 51% over 2018-2040 across its two main scenarios (the 2020 publication did not include a New Policies Scenario). In the Stated Policies Scenario, the report's central scenario, annual nuclear generation increases by 729 TWh (27%) between 2018 and 2040, requiring an increase in capacity of 59 GW, or 14%. In the report's Sustainable Development Scenario, nuclear generation increases by 1610 TWh (60%) over the same period, requiring capacity growth of about 179 GW, or 43%.

WEO 2019 presents electricity generation growth of between 51% and 67% over 2017-2040 across its three scenarios. In the Stated Policies Scenario, the report's central scenario, annual nuclear generation increases by 839 TWh (32%) between 2017 and 2040, requiring an increase in capacity of 69 GW, or 17%. In the report's Sustainable Development Scenario nuclear generation increases by 1773 TWh (67%) over the same period, requiring capacity growth of about 188 GW, or 46%.

WEO 2018 presents electricity generation growth of between 49% and 72% over 2016-2040 across its three scenarios. In the New Policies Scenario, the report's central scenario, annual nuclear generation increases by 1121 TWh (43%) between 2016 and 2040, requiring an increase in capacity of about 100 GW, or 25%. In the report's Sustainable Development Scenario nuclear generation increases by 2355 TWh (90%) over the same period, requiring capacity growth of about 265 GW, or 65%.

WEO 2017 presents electricity generation growth of between 48% and 75% over 2015-2040 across its three scenarios. In the New Policies Scenario, nuclear generation increases by 1273 TWh (50%) between 2015 and 2040, requiring an increase in capacity of about 100 GW, or 25%. In the report's Sustainable Development Scenario,

nuclear generation increases by 2774 TWh (108%) over the same period, requiring capacity growth of about 300 GW, or 75%.

WEO 2016 presents electricity generation growth of between 43% and 78% over 2014-2040 across its three scenarios. In the New Policies Scenario, nuclear generation increases by 1997 TWh (78%) between 2014 and 2040, requiring an increase in capacity of about 200 GW, or 45%. In the report's 450 Scenario, nuclear generation increases by 3566 TWh (141%) over the same period, requiring capacity growth of about 300 GW, or 95%.

WEO 2015 presents electricity generation growth of between 45% and 84% over 2013-2040 across its three scenarios. In the New Policies Scenario, nuclear generation increases by 2128 TWh (86%) between 2013 and 2040, requiring an increase in capacity of about 220 GW, or 55%. In the report's 450 Scenario, nuclear generation increases 3765 TWh (152%) over the same period, requiring capacity growth of about 450 GW, or 115%.

In June 2015 the IEA's *World Energy Outlook 2015 Special Report on Energy and Climate Change* was published, which "has the pragmatic purpose of arming COP21 negotiators with the energy sector material they need to achieve success in Paris in December 2015". It outlines a strategy to limit global warming to 2°C, but is very much focused on renewables.

The report recommended a series of measures including increasing energy efficiency, reducing the use of inefficient coal-fired power plants, increasing investment in renewables, reducing methane emissions, and phasing out fossil fuels subsidies. Half of the additional emissions reductions in its 450 Scenario come from decarbonisation efforts in power supply, driven by high carbon price incentives. In this scenario, an additional 245 GWe of nuclear capacity is built by 2040 compared with a moderate 'Bridge' option. The IEA acknowledges that nuclear power is the second-biggest source of low-carbon electricity worldwide after hydropower and that the use of nuclear energy has avoided the release of 56 billion tonnes of CO_2 since 1971, equivalent to almost two years of global emissions at current rates. The report suggests that intended nationally determined contributions (INDCs) submitted by countries in advance of COP21 will have trivial effect, and its purpose is clearly to suggest more ambitious emission reduction targets in its 'Bridge' scenario.

While the report confirms that nuclear energy needs to play an important role in reducing greenhouse gas emissions, it projects nuclear capacity of only 542 GWe (38% increase), producing 4005

TWh, by 2030 in its main 'Bridge' scenario. Most of the new nuclear plants are expected to be built in countries with price-regulated markets or where government-owned entities build, own, and operate the plants, or where governments act to facilitate private investment.

WEO 2014 had a special focus on nuclear power, and extended the scope of scenarios to 2040. In its New Policies Scenario, installed nuclear capacity growth is 60% through 543 GWe in 2030, and to 624 GWe in 2040 out of a total of 10,700 GWe, with the increase concentrated heavily in China (46% of it), plus India, Korea, and Russia (30% of it together) and the USA (16%), countered by a 10% drop in the EU. Despite this, the percentage share of nuclear power in the global power mix increases to only 12%, well below its historic peak. The 450 Scenario gives a cost-effective transition to limiting global warming assuming an effective international agreement in 2015, and this brings about a more than doubling of nuclear capacity to 862 GWe in 2040, while energy-related CO_2 emissions peak before 2020 and then decline. In this scenario, almost all new generating capacity built after 2030 needs to be low-carbon.

"Despite the challenges it currently faces, nuclear power has specific characteristics that underpin the commitment of some countries to maintain it as a future option," it said. "Nuclear plants can contribute to the reliability of the power system where they increase the diversity of power generation technologies in the system. For countries that import energy, it can reduce their dependence on foreign supplies and limit their exposure to fuel price movements in international markets."

Carbon dioxide emissions from coal use level off after 2020 in the New Policies Scenario, though CCS is expected to be negligible before 2030. CO_2 emissions from gas grow strongly to 2040.

WEO 2014 expressed concern about subsidies to fossil fuels, "which encourage wasteful consumption" and totalled $548 billion in 2013, over half of this for oil. Ten countries account for almost three-quarters of the world total for fossil-fuel subsidies, five of them in Middle East (notably Iran and Saudi Arabia) or North Africa where much electricity is generated from oil, and where nuclear power plants and renewables would be competitive, but for those subsidies. The report advocates ensuring "that energy prices reflect their full economic value by introducing market pricing and removing price controls." Renewables subsides in 2013 are put at $121 billion and rising, $45 billion of this being solar PV. Geographically this is $69 billion for EU and $27 billion in USA. The

report was unable to assign a figure for nuclear subsidies, which at present don't exist.

Following the Fukushima accident, ***WEO 2011*** New Policies Scenario had a 60% increase in nuclear capacity to 2035, compared with about 90% the year before. "Although the prospects for nuclear power in the New Policies Scenario are weaker in some regions than in [*WEO 2010*] projections, nuclear power continues to play an important role, providing base-load electricity. ... Globally, nuclear power capacity is projected to rise in the New Policies Scenario from 393 GW in 2009 to 630 GW in 2035, around 20 GW lower than projected last year." In this scenario the IEA expected the share of coal in total electricity to drop from 41% now to 33% in 2035. *WEO 2011* also included a "Low Nuclear Case (which) examines the implications for global energy balances of a much smaller role for nuclear power. Its effect would be to "increase import bills, heighten energy security concerns and make it harder and more expensive to combat climate change."

IEA: Net Zero by 2050

Net Zero by 2050, released in May 2021, outlines a possible roadmap for the global energy sector to achieve net zero emissions by mid-century. In the roadmap, the amount of energy provided by nuclear nearly doubles between 2020 and 2050. To achieve this, new capacity additions reach 30 GW per year in the early 2030s.

The amount of energy consumption that is in the form of electricity increases from about 20% today to about 50% by 2050. Whilst absolute supply from nuclear increases, its relative contribution to the electricity mix decreases from about 10.5% in 2020 to about 8% in 2050.

The report warned: "Failing to take timely decisions on nuclear power ... would raise the costs of a net-zero emissions pathway and add to the risk of not meeting the goal."

IEA: Energy Technology Perspectives

Energy Technology Perspectives (ETP) 2020 from the IEA says that, with a rising share of electricity in final consumption, "the technological transformation of the power generation sector is a central element of the clean energy transition. Decarbonisation drives down the carbon intensity of electricity generation: it falls from 463 grams of CO_2 per kilowatt-hour in 2019 to below zero in

net terms around 2055." However, in its Sustainable Development scenario with a threefold increase in total power generation, it projects only 780 GWe nuclear providing 8% in 2070. To support its projection of 84% from renewables, it projects 2100 GWe of utility-scale storage including 300 GWe pumped hydro, the rest being mainly by batteries with an average discharge duration of five hours.

ETP 2017 analyses various energy sector development paths to 2060 and notes: "In the power sector, renewables and nuclear capacity additions supply the majority of demand growth... Innovative transportation technologies are gaining momentum and are projected to increase electricity demand." Rising living standards will increase demand. "Nuclear power benefits from the stringent carbon constraint in the [Beyond 2 Degrees Scenario], with its generation share increasing to 15% by 2060 and installed capacity compared with today more than doubling to 1062 GWe by 2060. Of this, 64% is installed in non-OECD countries, with China alone accounting for 28% of global capacity... Achieving this long-term deployment level will require construction rates for new nuclear capacity of 23 GWe per year on average between 2017 and 2060." (p295)

ETP 2016 focused on the urban environment, since cities "represent almost two-thirds of global primary energy demand and account for 70% of carbon emissions in the energy sector." Its 2DS scenario to 2050 gives a major role to renewables in reducing emissions and much less to nuclear power, while maintaining optimism on CCS. For electricity, generation is almost completely decarbonized by 2050, achieved with 67% renewables including hydro (30% solar PV and wind), 12% coal and gas with CCS, and 16% nuclear (about 7000 TWh, from 914 GWe). Electric vehicles will account for 450 TWh. However, it notes that CCS development is languishing and "is not on a trajectory to meet the 2DS target of 540 Mt CO_2 being stored per year in 2025," and in 2015 "only 7.5 Mt/yr (27%) of the captured CO_2 is being stored with appropriate monitoring and verification."

ETP 2015 developed the earlier scenarios. In the main 2DS scenario, the share of fossil fuels in global primary energy supply drops by almost half – from 80% in 2011 to just over 40% in 2050. Energy efficiency, renewables and CCS make the largest contributions to global emissions reductions under the scenario. Under the 2DS scenario, some 22 GWe of new nuclear generating capacity must be added annually by 2050.

Launching ETP 2015, the IEA said: "A concerted push for clean-energy innovation is the only way the world can meet its climate goals," and that governments should help boost or accelerate this transformation."

ETP 2014 developed the ETP 2012 scenarios. In the 2DS one which is the main focus, some 22 GWe of new nuclear generating capacity must be added annually by 2050. However, the IEA notes that global nuclear capacity "is stagnating at this time" and by 2025 will be 5% to 25% below needed levels, "demonstrating significant uncertainty." It suggests that the high capital and low running costs of nuclear create the need for policies that provide investor certainty.

The IEA estimated that an additional $44 trillion in investment was needed in global electricity systems by 2050. However, it says that this represents only a small portion of global GDP and is offset by over $115 trillion in fuel savings.

Launching the ETP 2014 report, the IEA executive director said: "Electricity is going to play a defining role in the first half of this century as the energy carrier that increasingly powers economic growth and development. While this offers opportunities, it does not solve our problems; indeed, it creates many new challenges."

European Commission

In December 2011 the European Commission (EC) published its ***Energy 2050 Roadmap***, a policy paper. This was very positive regarding nuclear power and said that nuclear energy can make "a significant contribution to the energy transformation process" and is "a key source of low-carbon electricity generation" that will keep system costs and electricity prices lower. "As a large scale low-carbon option, nuclear energy will remain in the EU power generation mix." The paper analysed five possible scenarios leading to the EU low-carbon energy economy goal by 2050 (80% reduction of CO_2 emissions), based on energy efficiency, renewables, nuclear power and carbon capture and storage (CCS). All scenarios show electricity will have to play a much greater role than now, almost doubling its share in final energy demand to 36%-39% in 2050. The EC high-efficiency scenario would reduce energy demand by 41% by 2050 (compared with 2005); the diversified supply technologies scenario would have a combination of high carbon prices, nuclear energy and introduction of CCS technologies; a high-renewables scenario suggests they might supply 75% of total energy supply by 2050; a "delayed CCS"

scenario has nuclear power playing a major role; and a low-nuclear power scenario had coal plants with CCS providing 32% of total energy (ie 82-89% of EU electricity). The highest percentage of nuclear energy would be in the delayed CCS and diversified supply technologies scenarios, in which it would account for 18% and 15% shares of primary energy supply respectively, ie 38-50% of EU electricity. Those scenarios also had the lowest total energy costs.

Related information

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